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Network Security And Forensics

Lab Session – 2

Q1. Study and demonstration of network Cable.
Forming of network cable CAT-6 cables

A **network cable** is a physical medium used to transmit data between computers, switches, routers, and other network devices. Among various types of network cables, **CAT-6 (Category 6)** cables are widely used in modern Local Area Networks (LANs) due to their high speed and reliability.

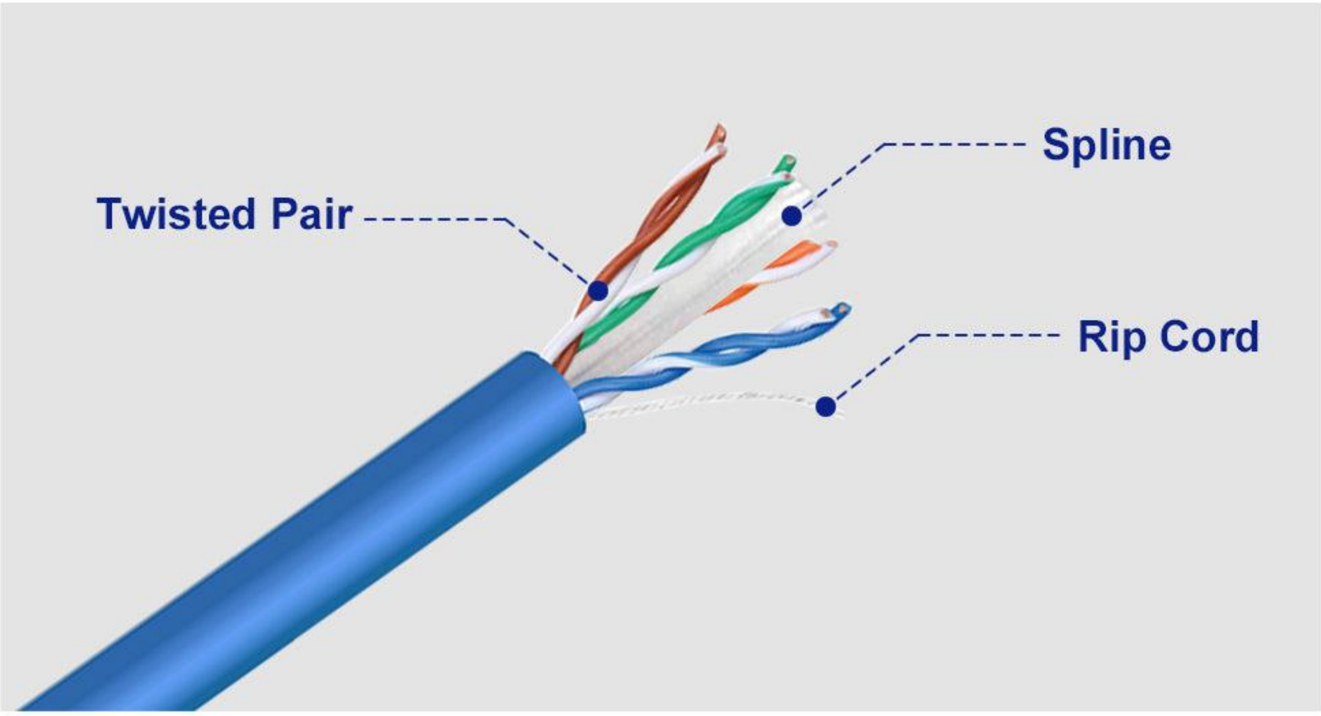
What is a Network Cable?

A network cable is a communication cable that carries data signals from one device to another in a wired network. It plays a critical role in ensuring stable, secure, and high-speed data transmission.

Types of Network Cables

1. **Twisted Pair Cable**
   - UTP (Unshielded Twisted Pair)
   - STP (Shielded Twisted Pair)
2. **Coaxial Cable**
3. **Fiber Optic Cable**

CAT-6 Network Cable



CAT-6 is an advanced Ethernet cable standard designed to support **high-speed data transmission up to 1 Gbps (1000 Mbps)** and frequencies up to **250 MHz**.

Features of CAT-6 Cable

- Supports high data rates
- Reduced crosstalk and interference
- Suitable for Gigabit Ethernet
- Better performance compared to CAT-5e
- Maximum recommended length: **100 meters**

Internal Structure

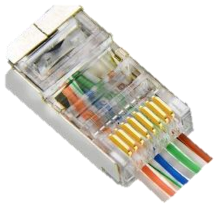
A CAT-6 cable contains:

- **4 twisted pairs (8 wires total)**
- Color-coded insulation
- Central spline (in some cables) to reduce interference
- Outer protective jacket

Forming (Crimping) of CAT-6 Network Cable

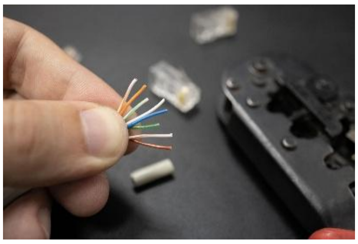
Required Tools

CAT-6 Ethernet cable & RJ-45 connectors



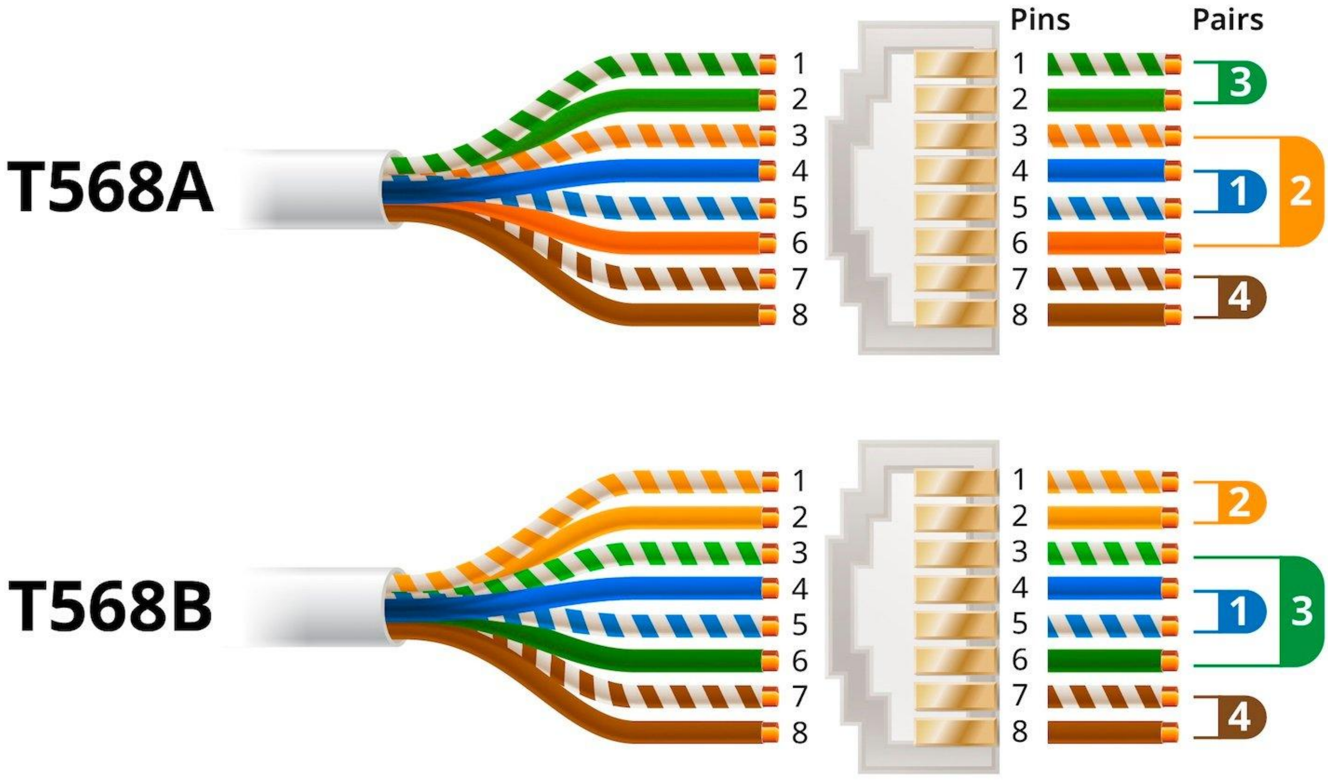
Crimping tool

Wire stripper / cutter



Wiring Standards

Two common wiring standards are used:



Steps to Form CAT-6 Network Cable (Short)

1. **Cut the cable** to the required length.
2. **Strip the outer jacket** (about 1–1.5 inches).
3. **Untwist and arrange wires** according to **T568B** (or T568A) standard.
4. **Trim the wires evenly** in a straight line.
5. **Insert wires into RJ-45 connector** (ensure correct order).
6. **Crimp the connector** using a crimping tool.
7. **Test the cable** using a LAN cable tester.

Q2. **Study of various internetworking devices**
Hub, Switches & Routers

Hub

A **hub** is a basic networking device that connects multiple computers in a network and sends data to all connected devices.

How it works?

- When a device sends data to the hub,
- The hub **broadcasts the data to every port**,
- All devices receive the data, but only the intended device accepts it.

Characteristics

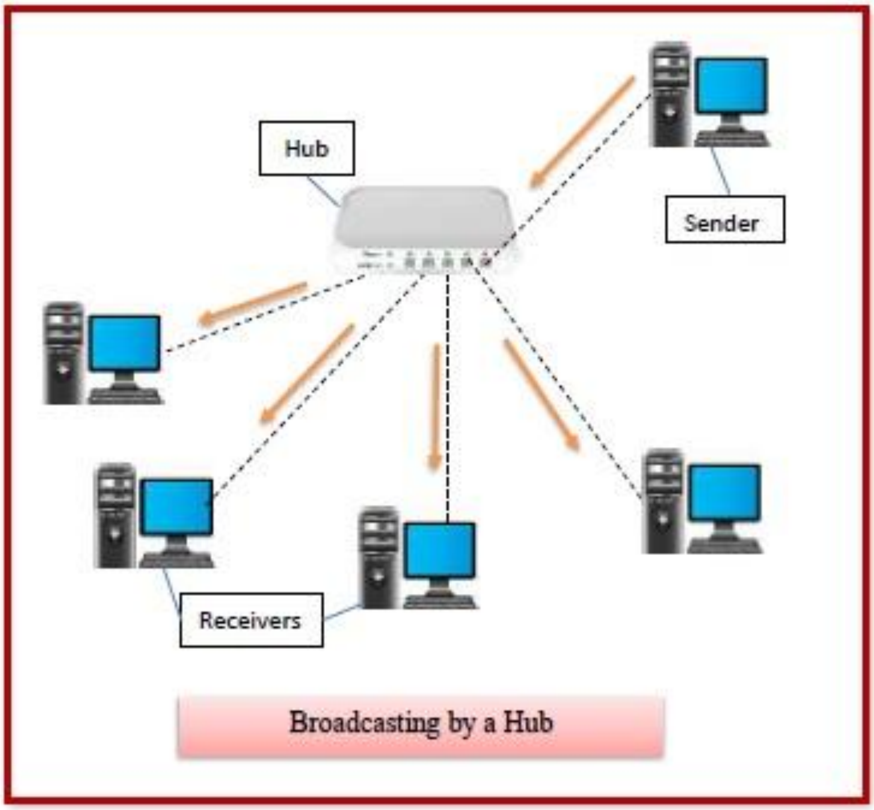
- Works at **Physical Layer (Layer 1)** of OSI model
- No intelligence or decision-making ability
- Cannot identify destination device
- Causes data collision

Advantages

- Easy to install and use
- Low cost

Disadvantages

- Poor performance
- No security
- Not used in modern networks



Switches

A switch is a networking device that connects multiple devices in a LAN and sends data only to the specific destination device.

How it works ?

- A switch learns MAC addresses of connected devices
- Stores them in a MAC address table
- Forwards data only to the correct port

Characteristics

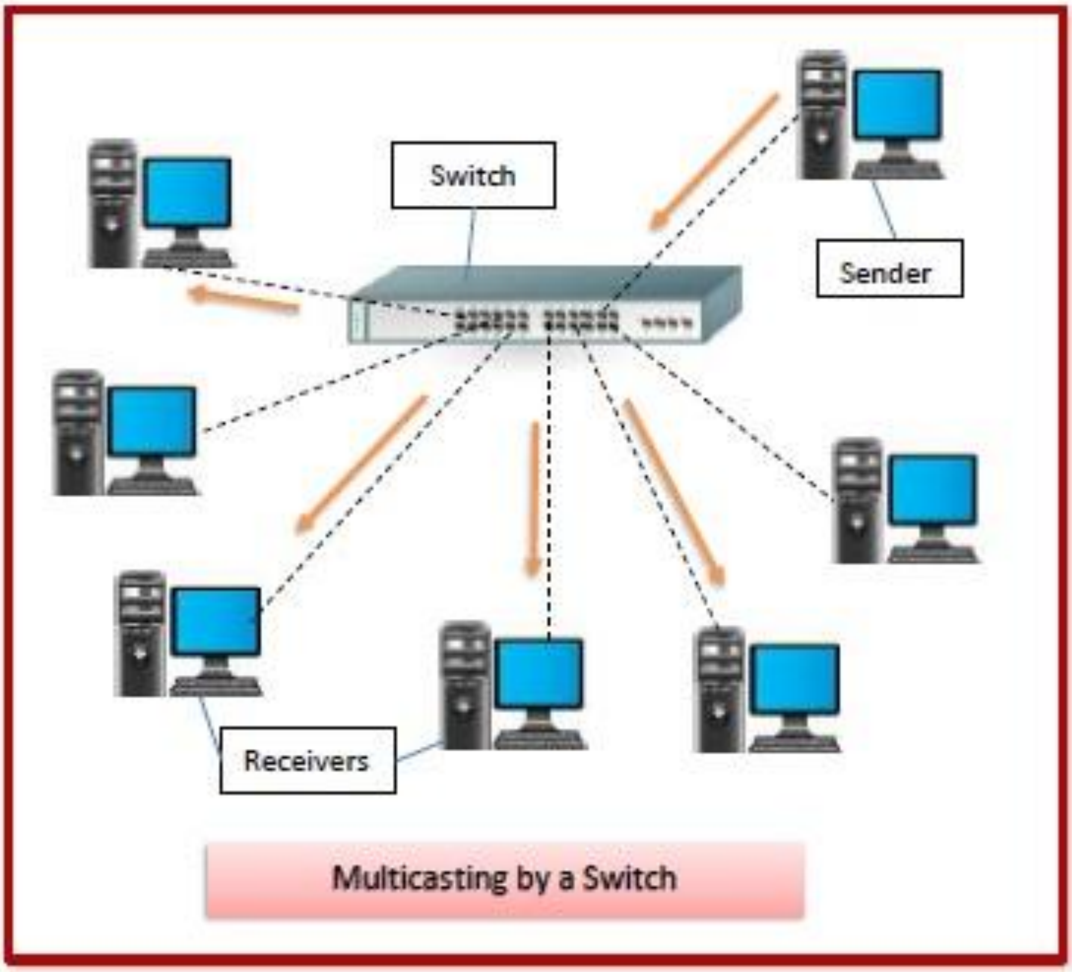
- Works at Data Link Layer (Layer 2)
- Reduces collisions
- Faster and smarter than hub

Advantages

- High speed and efficiency
- Better security than hub
- Less network congestion

Disadvantages

- Costlier than hub
- Limited to local networks



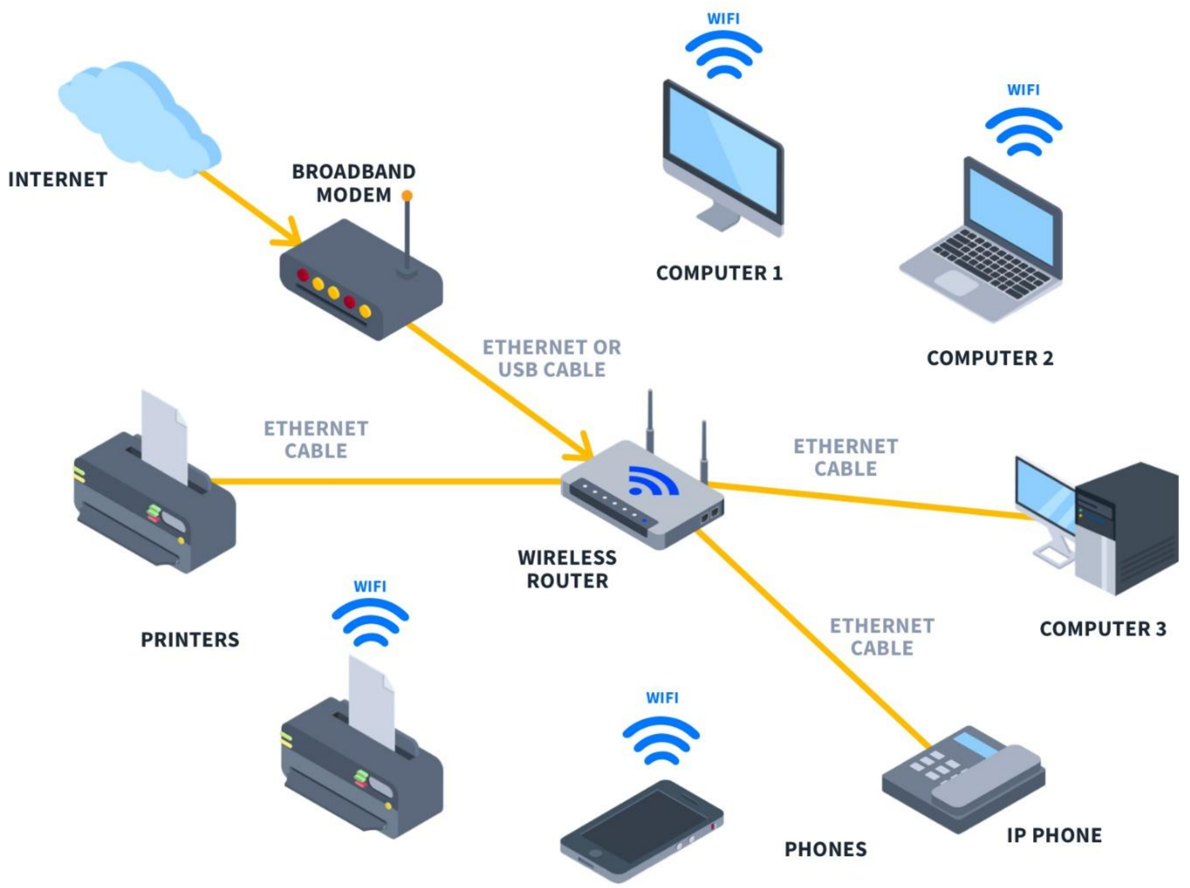
Routers
A **router** is a networking device that connects different networks and routes data packets using IP addresses.

Working
- Receives data packets
- Checks **destination IP address**
- Uses **routing table** to find the best path
- Forwards data to the correct network

Characteristics
- Works at **Network Layer (Layer 3)**
- Connects LAN to WAN or Internet
- Supports routing protocols

Advantages
- Enables internet connectivity
- Provides security using firewall and NAT
- Efficient traffic management

Disadvantages
- Expensive
- Complex configuration



Q3. Demonstration of IP Change and IP Scanner

An IP address (Internet Protocol address) uniquely identifies a device on a network. Changing the IP address and scanning IPs are common network administration and troubleshooting tasks used to manage devices, detect active hosts, and enhance security.

Demonstration of IP Address Change

Objective

To change the IP address of a computer manually and verify the change.

Methods of Changing IP Address

1. Automatic IP (DHCP)
The IP address is assigned automatically by a DHCP server (usually a router).

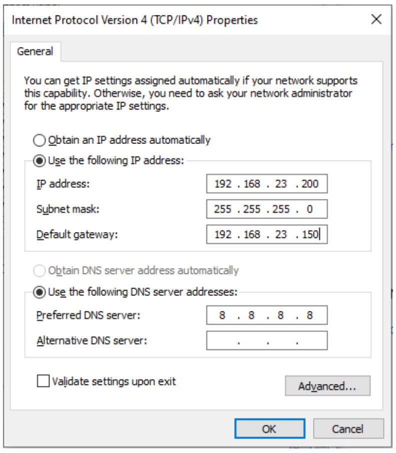
2. Manual IP (Static IP)
The IP address is manually assigned by the user.

Steps to Change IP Address (Windows – Static IP)

1. Open **Control Panel** → **Network and Internet** → **Network and Sharing Center**
2. Click **Change adapter settings**
3. Right-click on **Ethernet / Wi-Fi** → **Properties**
4. Select **Internet Protocol Version 4 (IPv4)** → Click **Properties**
5. Select **Use the following IP address**
6. Enter:
   - IP Address (e.g., 192.168.1.10)
   - Subnet Mask (e.g., 255.255.255.0)
   - Default Gateway (e.g., 192.168.1.1)
7. Click **OK** to save settings

Verification

Open **Command Prompt** and type:
ipconfig



Demonstration of IP Scanner

Objective

To scan a network and identify active devices using an IP scanner.

What is an IP Scanner?

An **IP scanner** is a tool that scans a range of IP addresses to detect:

- Active devices
- IP addresses
- MAC addresses
- Hostnames
- Open ports (in advanced scanners)

Steps to Use an IP Scanner

1. Open the IP scanner tool
2. Enter the **IP range** (e.g., 192.168.1.1 – 192.168.1.254)
3. Click **Scan**
4. The scanner displays:
   - Active IPs
   - Device names
   - MAC addresses

Result

All devices connected to the same network are listed with their IP details.

Conclusion

The demonstration of IP address change helps understand network configuration and management. IP scanners are essential tools for detecting active devices and maintaining network security. Both concepts are fundamental in networking and cybersecurity.